

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 1

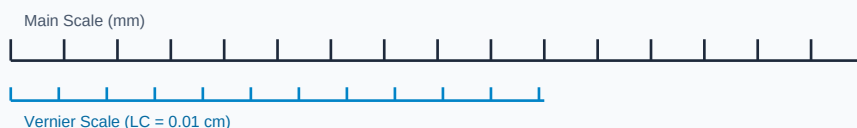
CHAPTER 5: MOTION OF SYSTEM OF PARTICLES AND RIGID BODIES

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 1
Chapter Name:	Motion of System of Particles and Rigid Bodies

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

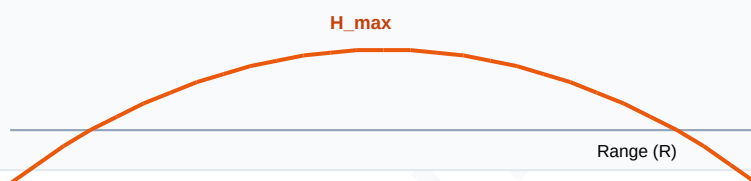
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Center of Mass	The point where the entire mass of a system of particles can be assumed to be concentrated.
Torque	The turning effect of a force about a point or axis of rotation, measured as force times distance.
Radius of Gyration	The perpendicular distance from the axis of rotation to a point where the mass of the rigid body is concentrated.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

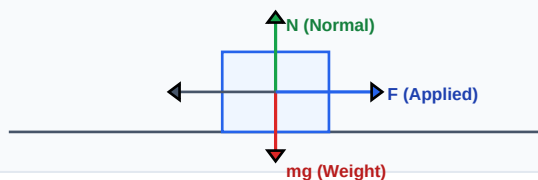
Kinematic Projectile Trajectory



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Center of Mass: $X_{cm} = (m_1 \cdot x_1 + m_2 \cdot x_2) / (m_1 + m_2)$
- Torque: $\tau = r \times F = I \cdot \alpha$
- Angular Momentum: $L = r \times p = I \cdot \omega$
- Parallel Axis Theorem: $I = I_{cm} + M \cdot d^2$ | Perpendicular: $I_z = I_x + I_y$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

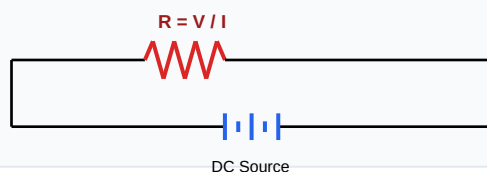
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Two masses of 2kg and 3kg are placed at coordinates $x=1\text{m}$ and $x=4\text{m}$. Find the center of mass coordinate.

Step-by-step Solution:

$$X_{\text{cm}} = (2 \cdot 1 + 3 \cdot 4) / (2 + 3) = (2 + 12) / 5 = 14 / 5 = 2.8 \text{ meters.}$$

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: State and prove the perpendicular axes theorem for planar laminar bodies.

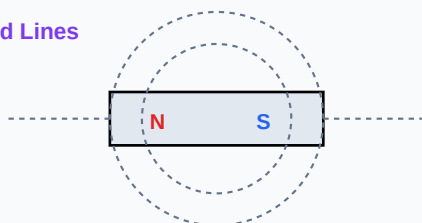
Analytical Solution Strategy:

Theorem states $I_z = I_x + I_y$.

Proof: For any particle mass m at (x,y) , $I_z = \sum [m * (x^2 + y^2)]$.

$I_z = \sum [m * x^2] + \sum [m * y^2] = I_y + I_x$, which matches definition of axes.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Define moment of inertia and state its SI unit.

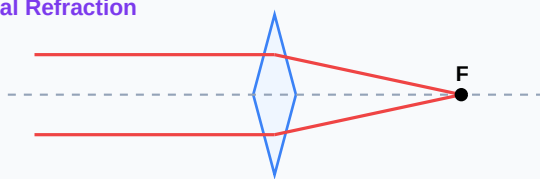
Q2 (2-Mark): Explain conservation of angular momentum with an example.

Q3 (2-Mark): Write the parallel axis theorem expression.

Q4 (5-Mark): Find the torque of force $F = 2i + 3j$ acting at distance $r = i - j$.

Q5 (5-Mark): Distinguish between center of mass and center of gravity.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Aviation Dynamics

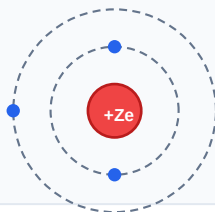
Helicopters balance rotor torque using tail rotors to maintain yaw stability.

Application Case: Gymnastics

Divers tuck their limbs in during jumps to reduce their moment of inertia, spinning faster.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

